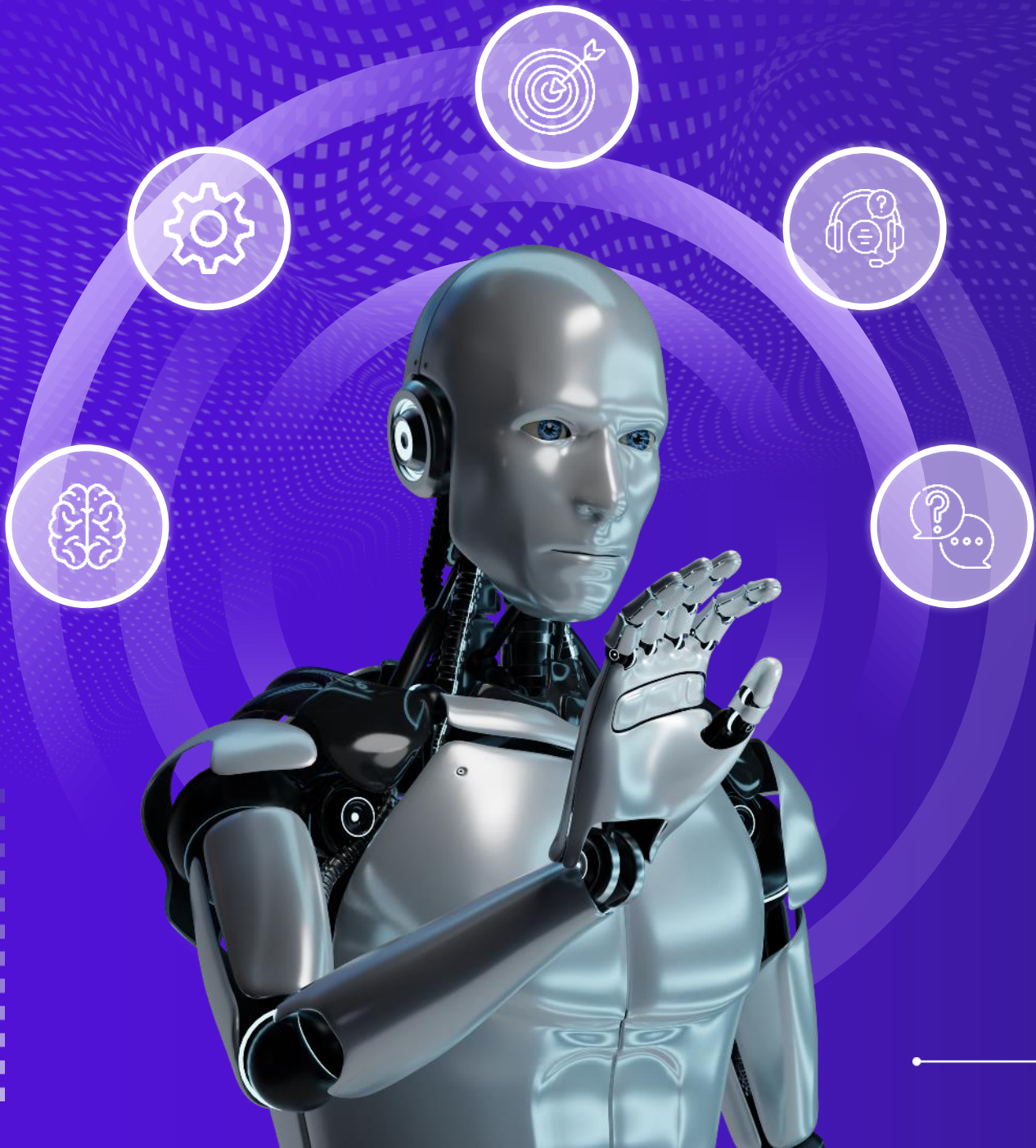




Agentic AI

Transforming Automation & Intelligence



What is Agentic AI

Agentic AI integrates **large language models (LLMs), machine learning, and enterprise automation** to create autonomous AI agents capable of **decision-making, goal-setting, problem-solving, and adapting to unstructured environments**, unlike rule-based AI.



Evolution of Agentic AI



Early AI Models

Focused on predefined rules and automation.



Machine Learning Era

AI began adapting and learning patterns.



Rise of Large Language Models (LLMs)

Enhanced contextual understanding.



Agentic AI Integration

Combining AI, automation, and real-time adaptability.

How Agentic AI Works



Understanding

AI processes data using LLMs and machine learning.



Goal-Setting

Identifies objectives and plans strategies.



Execution

Automates tasks with minimal human input.



Learning

Continuously improves through interactions.

Benefits Of Agentic AI



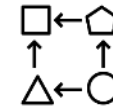
Efficiency

Reduces manual work and automates complex tasks, improving productivity.



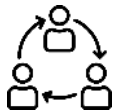
Scalability

AI agents can adapt to business growth and evolving demands.



Adaptability

Responds dynamically to real-time changes and unstructured environments.



Collaboration

Works alongside humans, enhancing decision-making and workflow efficiency.



Cost Reduction

Lowers operational expenses by automating repetitive and resource-intensive processes.



Continuous Learning

Improves performance over time through data-driven insights and machine learning.

Agentic AI Vs. Generative AI

Agentic AI

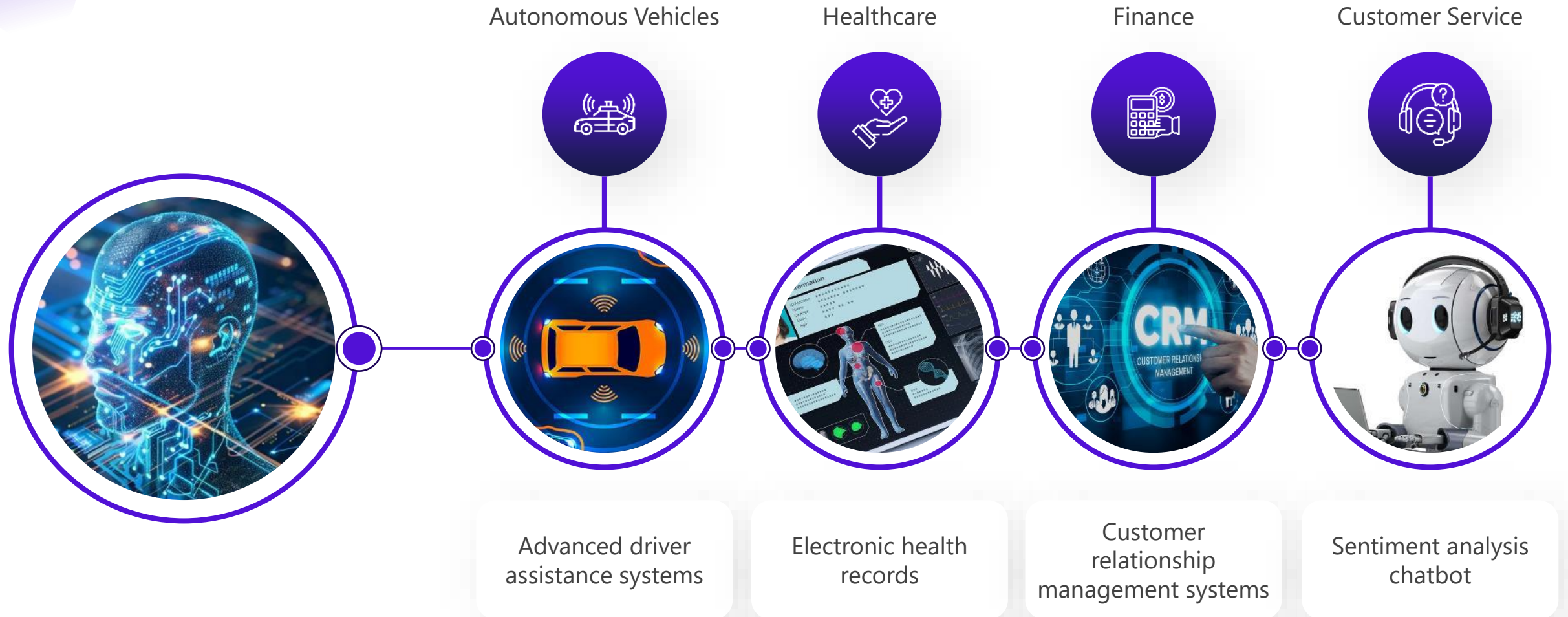
- ➔ Autonomous
- ➔ Goal-Oriented
- ➔ Real-time Environment
- ➔ Utilize Tools and APIs
- ➔ Continuous Adaptation
- ➔ Autonomous Systems
- ➔ Independent Decision-Making

Vs

Generative AI

- ➔ Creative
- ➔ Output-Oriented
- ➔ Content-Based
- ➔ No external tools
- ➔ No adaptation from real-time
- ➔ Content Creation
- ➔ Make decision based on input patterns

Applications Of Agentic AI



Use Cases of Agentic AI & Automation



Automated Data Analysis

AI interprets trends in real-time.



Supply Chain Optimization

Predictive analytics improve logistics.



IT & Security

AI-driven cybersecurity threat detection.



Self-Healing Systems

AI resolves technical issues autonomously.





Risks & Challenges of Agentic AI

Risks

- ✓ Bias & Fairness Issues
- ✓ Security Threats
- ✓ Loss of Human Control

Challenges

- ✓ Ethical Concerns
- ✓ Regulatory & Compliance Issues
- ✓ Accountability & Transparency

Best Practices For Implementing



01

Ethical AI Development

Ensure transparency and fairness in decision-making.

02

Robust Security Measures

Prevent unauthorized access and cyber threats.

03

Continuous Learning

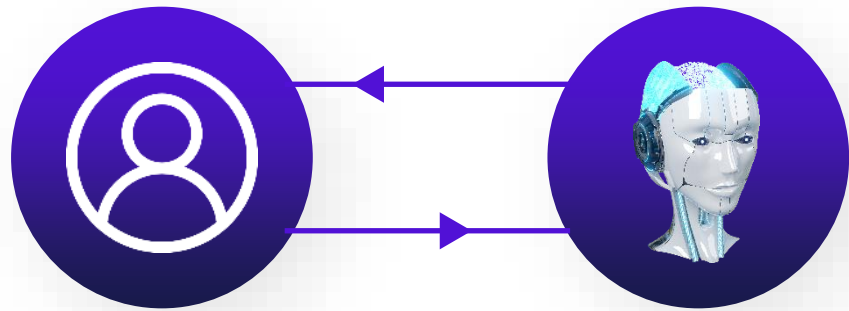
Enable AI to adapt and improve with new data.

04

Human Oversight

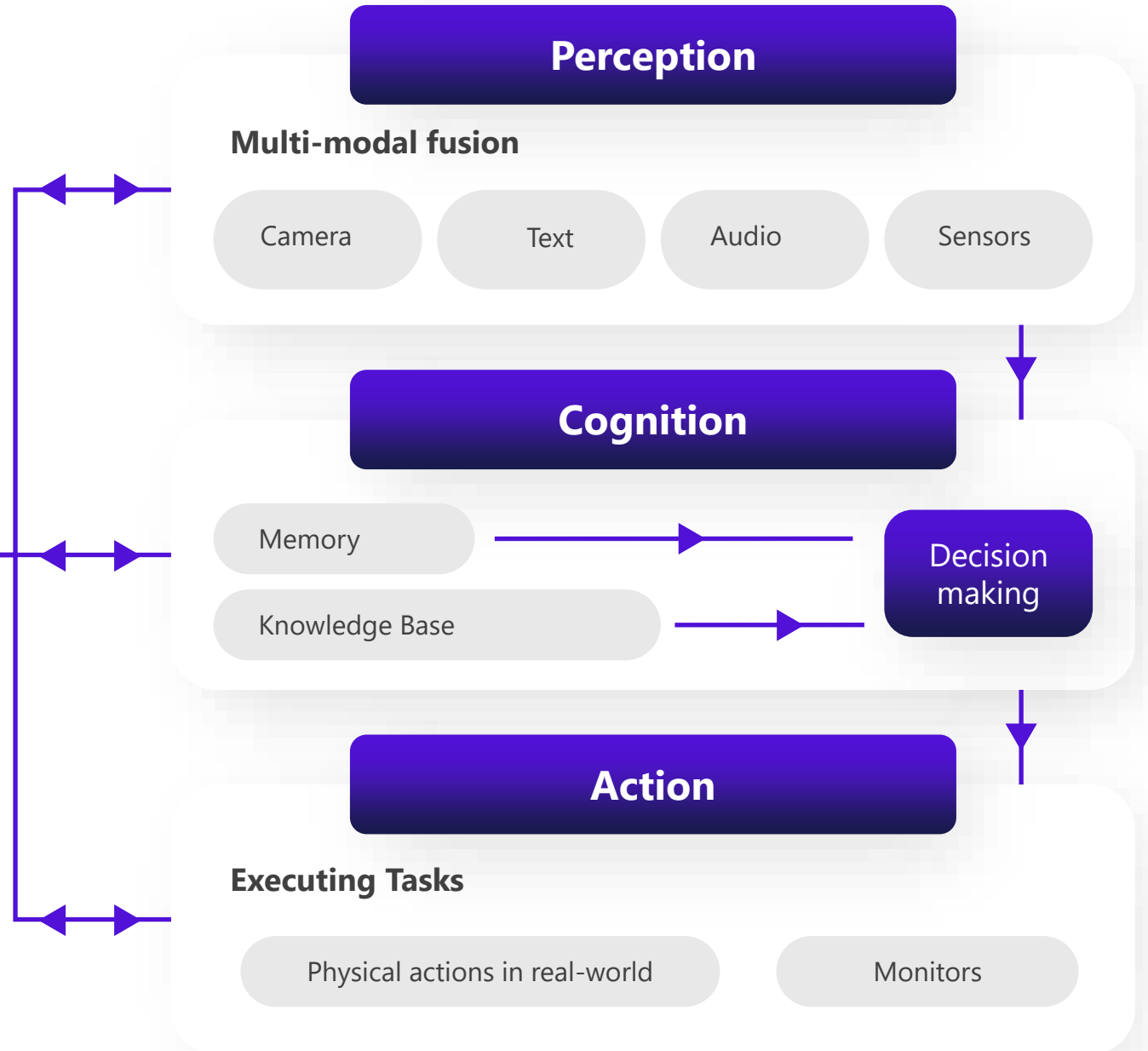
Maintain control to align AI decisions with business values.

Agentic AI System



User Request

AI Agent



The Future of Agentic AI

The future of Agentic AI involves autonomous decision-making, human collaboration, and enhanced AI security.



Autonomous AI Agents

AI will make more independent decisions and perform complex tasks.



AI & Human Collaboration

AI will work alongside humans, enhancing efficiency and innovation.



Next-Gen AI Security

Strengthening AI safety to prevent risks and ensure ethical use.

Conclusion & Key Takeaways

Agentic AI is revolutionizing **automation beyond simple processes**. Businesses must adapt as AI shapes the future of work. Ethical **AI development** is essential for responsible use, and AI will continue evolving, driving innovation and transformation.



Large Language Models for AI

Developer / Organization	Country / Origin	Models Provided
OpenAI	us USA	GPT-5, GPT-4.5, GPT-o1, GPT-o3; (some free open-weight: gpt-oss-20b, gpt-oss-120b)
Meta (formerly Facebook AI)	us USA	LLaMA-2 (7B, 13B, 70B), LLaMA-3 (8B, 70B, 405B), Code LLaMA (various)
Microsoft	us USA	Phi-3 (mini/small/medium), Phi-4 models; research & internal models
xAI (Elon Musk)	us USA	Grok-1 / Grok-1.5 / Grok-2 / Grok-3; Grok Vision variants
Google (DeepMind / Google AI)	us USA	Gemini series (e.g., Gemini 2.5), Gemma open models (Gemma 1/2/3, vision / domain variants)
Anthropic	us USA	Claude 4.1, Claude 3.7 Sonnet, Claude 3.5
Alibaba Cloud	cn China	Qwen series (Qwen 1.5, Qwen 2.5, Qwen 3, multimodal / coded versions)
DeepSeek AI	cn China	DeepSeek-V2/V3, DeepSeek-R1, DeepSeek-Coder
Mistral AI	FR France	Mistral-7B, Mistral Large 2, Mixtral (Mixture-of-Experts)
Technology Innovation Institute (TII)	AE UAE	Falcon series (Falcon 180B, Falcon 40B, etc.)
Databricks / Mosaic ML	us USA	DBRX (132B mixture-of-experts)
Stability AI	GB UK	StableLM family (StableLM 1.6B, 12B), other multimodal models
EleutherAI (Community)	🌐 Global	GPT-NEOX 20B, GPT-J 6B
LAION / OpenAssistant	🌐 Global	OpenAssistant (30B)
Zhipu AI / Z.ai	cn China	GLM-4.5 / GLM-4.5 Air (MoE reasoning models)
01.AI / Yi Models	KR South Korea	(Reported open science models)
TigerBot / LMSYS (Community)	🌐 Global	TigerBot family (7B, 13B, 70B, 180B)
BLOOM (BigScience)	🌐 Global	BLOOM 176B, multilingual open models

Free vs Paid Models ***“Free”** = open-weight or locally runnable

***“Paid”** = requires cloud subscription / API usage

Model Family	Developer	Free	Paid	Notes
Phi-3-mini / Phi-3	Microsoft	✓	✗	Best lightweight free model for labs
LLaMA-3 (various sizes)	Meta	✓	✗	Open weights, local use
Mistral-7B / Large2	Mistral	✓	✗	Strong free performance
Gemma-2 / 9B / 27B	Google	✓	✗	Free / open models
Qwen Series	Alibaba	⚠	⚠	Some free (small sizes), some paid larger
DeepSeek-LLM / R1 / V3	DeepSeek AI	✓	✗	Free open source family
Falcon (40B / 180B)	TII	✓	✗	Free open large models
StableLM	Stability AI	✓	✗	Free voice + text
OpenAssistant	LAION	✓	✗	Community open model
GPT (GPT-4 / 5 / o-series)	OpenAI	✗	✓	Classic paid cloud models
Claude (Anthropic)	Anthropic	✗	✓	Paid API
Grok (all versions)	xAI	✗	Partial	Paid / X Premium - partial access
Gemini (cloud)	Google	✗	✓	Cloud only (Gemini /*β)

Best Models for Generative vs Agentic Tasks

Model Family	Best for Generative Tasks	Best for Agentic Tasks	Notes
GPT-4.5 / GPT-5 / o-series (OpenAI)	★★★★	★★★★	Best general performance, paid
Claude 3.7 / 4.1 (Anthropic)	★★★★	★★★	Strong for safe generation
Grok-3 (xAI)	★★★	★★★	Good text + real-time reasoning
LLaMA-3 (Meta)	★★★★	★★★	Strong gen, ok reasoning
Phi-3-mini / Phi-3 (Microsoft)	★★★★	★★★★	Excellent reasoning, instruct
Mistral-7B / Large2	★★★★	★★★	Balanced generative
Gemma-2 / 9B / 27B (Google)	★★★★	★★★	Strong generative
Qwen 2.5 / 3.0 (Alibaba)	★★★★	★★★★	Balanced for both goals
DeepSeek-LLM / R1 / V3	★★★	★★★★	R1/V3 strong for agentic
Falcon (40B / 180B)	★★★★	★★★	Good gen with long context
StableLM (Stability AI)	★★★	★★	Good gen, weaker reasoning
OpenAssistant	★★★	★★	Community gen model

Best Models for Generative vs Agentic Tasks

Framework	Category	Primary Use	Best For	Typical Projects
LangChain	Hybrid (Gen + Agentic)	LLM orchestration	Prompt chaining, tools, agents	Chatbots, RAG, simple agents
LangGraph	Agentic AI	State-based agents	Multi-step reasoning, workflows	Autonomous agents, planners
CrewAI	Agentic AI	Multi-agent systems	Role-based agents	Research agents, team agents
AutoGen (Microsoft)	Agentic AI	Conversational agents	Agent-to-agent interaction	Autonomous decision systems
Haystack	Generative AI	NLP pipelines	Search + generation	RAG, QA systems
LlamaIndex	Generative AI	Data → LLM interface	Knowledge grounding	Document QA, RAG
Semantic Kernel	Agentic AI	Planning + tools	Enterprise agents	Task automation
DSPy	Generative AI	Prompt optimization	Structured prompting	Reliable LLM pipelines
Flowise	Generative AI	No-code LLM apps	Rapid prototyping	Chatbots, RAG
PromptFlow	Generative AI	LLM workflows	Evaluation & deployment	Production GenAI apps
Rasa	Agentic AI	Dialogue management	Stateful agents	AI assistants
Griptape	Agentic AI	Tool-driven agents	Secure agent workflows	Enterprise automation
OpenAI Assistants API	Agentic AI	Tool-calling agents	Hosted agents	SaaS copilots
IBM Watsonx.ai	Hybrid	Enterprise GenAI	Governance + agents	Regulated AI apps
Hugging Face Transformers	Generative AI	Model usage	Model fine-tuning	Text, vision, audio AI
Ray / Ray AIR	Agentic AI	Distributed agents	Scalable AI systems	Large agent systems

Agentic AI Frameworks

Agent Frameworks (High-Level Abstractions)

Framework	Key Features	Ideal For
LangChain	Tool integration, LLM agents, memory, planning	General-purpose autonomous agents
Autogen (by Microsoft)	Multi-agent collaboration, structured conversations	Research, multi-agent systems
CrewAI	Role-based multi-agent orchestration	Teams of AI agents, simulations
Haystack	Modular pipeline for RAG, with agent support	Question answering, document agents
AutoGPT / BabyAGI	Fully autonomous LLM agents with memory and planning	End-to-end task automation (experimental)
Google ADK	Simulated/task-completing agents (web, software, robotics)	Less documentation, mostly research use